

91262



NEW ZEALAND QUALIFICATIONS AUTHORITY
MANA TOHU MĀTAURANGA O AOTEAROA

2

DRD

SUPERVISOR'S USE ONLY

mm worked solutions
mm thinking
mm marking

Level 2 Mathematics and Statistics, 2014

91262 Apply calculus methods in solving problems

2.00pm Wednesday 19 November 2014
Credits: Five

Achievement	Achievement with Merit	Achievement with Excellence
Apply calculus methods in solving problems.	Apply calculus methods, using relational thinking, in solving problems.	Apply calculus methods, using extended abstract thinking, in solving problems.

Check that the National Student Number (NSN) on your admission slip is the same as the number at the top of this page.

You should attempt ALL the questions in this booklet.

Make sure that you have Resource Sheet L2-MATHF.

Show ALL working.

If you need more space for any answer, use the page(s) provided at the back of this booklet and clearly number the question.

Check that this booklet has pages 2–19 in the correct order and that none of these pages is blank.

YOU MUST HAND THIS BOOKLET TO THE SUPERVISOR AT THE END OF THE EXAMINATION.

TOTAL

ASSESSOR'S USE ONLY

QUESTION ONE

- (a) A function g is given by $g(x) = x^3 - 4x + 5$.

Find the gradient of the graph of g at the point where $x = 2$.

gradient
→ diff.

$$g'(x) = 3x^2 - 4$$

$$g'(2) = 3(2)^2 - 4$$

$$= 8$$

(a) derivative found
AND gradient found.

- (b) For a function f

$$f'(x) = 3x^2 + 4x - 1$$

The graph of f passes through the point $(2, 5)$.

$f'(x) \rightarrow f(x)$
anti-diff.

(c)

Find the equation of the function f .

$$f(x) = x^3 + 2x^2 - x + C$$

$$(2, 5)$$

$$5 = (2)^3 + 2(2)^2 - 2 + C$$

$$5 = 8 + 8 - 2$$

$$5 = 14 + C$$

$$C = -9$$

$$f(x) = x^3 + 2x^2 - x - 9$$

(a) anti-derivative
AND function found

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$$5 = 14 + c$$

$$c = -9$$

$$f(x) = x^3 + 2x^2 - x - 9$$

(a) anti-derivative
AND function found



- (c) A diver dives into a pool. The depth d metres that she reaches t seconds after she hits the water is given by

$$d(t) = 1.25t^2 - 4t$$

Find the greatest depth that she reaches.

positive quad.
greatest depth
= minimum.
min $\rightarrow d'(t) = 0$.

$$d'(t) = 2.5t - 4$$

$$2.5t - 4 = 0$$

(a) derivative
found AND
equated to 0
OR $t = 1.6$

$$2.5t = 4$$

$$t = \frac{4}{2.5} = 1.6$$

when $t = 1.6$

$$d(1.6) = 1.25(1.6)^2 - 4(1.6) \\ = -3.2$$

greatest depth is 3.2m
below the water which
occurs 1.6 sec after
she hits the water.

(m) time for min AND
depth found.

- (d) The graph of $f(x) = -x^2 + kx - 5$ has a turning point at (3,4).

Find the gradient of the function at the point where $x = 4$.

$\rightarrow f'(x) = 0$ at TP

$\rightarrow$ shift

(3,4)
(4,-) $\rightarrow$ grad?
neg quad

$$f'(x) = -2x + k$$

tp at (3,4)

$$0 = -2(3) + k$$

$$k = 6$$

$$f'(x) = -2x + 6$$

(a) derivative found
AND $k = 6$ found

$$f'(4) = -2(4) + 6$$

$$f'(4) = -2$$

(m) gradient found.

(e) $g(x) = \frac{2x^3}{3} + \frac{3x^2}{2} - 20x + 4$

the cubic



For what values of x is g a decreasing function?

You must use calculus in finding your solution.

$$g'(x) = 2x^2 + 3x - 20$$

$$\text{min / max} \rightarrow g'(x) = 0$$

$$g'(x) = 2x^2 + 3x - 20 = 0$$

$$x = 2.5 \text{ or } x = -4$$

(a) derivative found
AND $= 0$ or < 0

when $x = 2.5$

$$g(x) = -26.2083$$

min

when $x = -4$

$$g(x) = 65\frac{1}{3}$$

max

(m) derivative found
and the x -values
of tp found

$g'(x)$ is a positive quad
-decreasing when
 $g'(x) < 0$.



region correctly
identified AND
justified

so $g(x)$ is decreasing
when

$$-4 < x < 2.5$$

- (f) During a fund-raising cycle ride, the distance s kilometres of a cyclist from a fixed point on his ride is modelled by the function

$$s(t) = 0.1875t^3 - 2.25t^2 + 21t + 0.5$$

where t is the time in hours since the cyclist passed the fixed point.

dist $\rightarrow$ speed
diff.

How far will the cyclist be from the fixed point when he reaches his minimum speed?

$$v(t) = s'(t) = 0.5625t^2 - 4.5t + 21$$

velocity



the quad
min when
grad = 0

↑ (a) velocity
eqn found

$$v'(t) = 1.125t - 4.5 = 0$$

$$1.125t = 4.5$$

$$t = 4$$

← (m) acceleration eqn
found AND set equal
to 0.

$$\text{when } t = 4 \quad s(4) = 0.1875(4)^3 - 2.25(4)^2 + 21(4) + 0.5$$

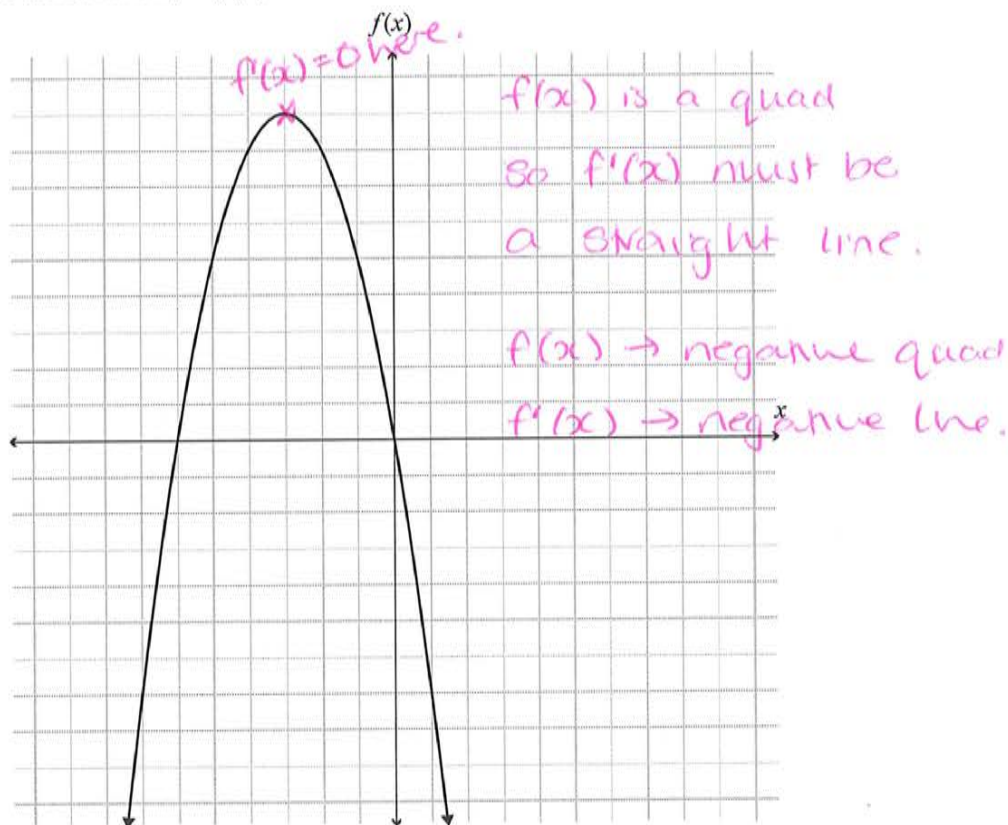
$$s(t) = 60.5$$

The cyclist will be 60.5m from the fixed point when he reaches his minimum speed.

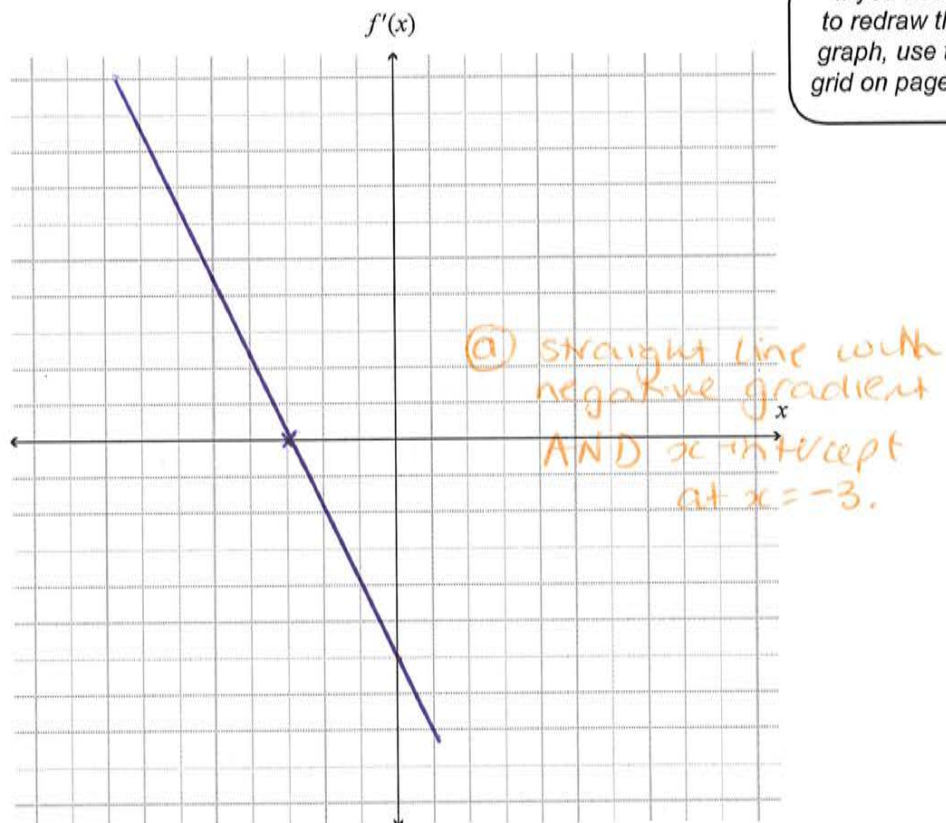
↑ (e) distance found.

QUESTION TWO

- (a) The graph of the function $y = f(x)$ is shown on the axes below.



On the axes below sketch the gradient function $y = f'(x)$.



If you need
to redraw this
graph, use the
grid on page 16

- (b) The graph of a function $y = g(x)$ passes through $(0,0)$, and its gradient function is $g'(x) = 2x - 5$.

Find the y -coordinate of the point on the curve where $x = 3$.

$$g(x) = x^2 - 5x + c$$

$$(0,0)$$

$$0 = 0 - 0 + c$$

$$c = 0$$

$$g(x) = x^2 - 5x$$

when $x = 3$

$$g(3) = (3)^2 - 5(3)$$

$$= 9 - 15$$

$$g(3) = -6$$

(a) anti-derivative found
AND y -coordinate found

- (c) Sione claims that the resale value of a car, t years after it is sold, can be modelled by the function

$$R = 150t^2 - 2250t + 38\,000, \text{ where } R \text{ is the resale value in dollars.}$$

How long after it is sold will the car's value be changing at a rate of $-\$150$ per year?

$$\frac{dR}{dt} = 300t - 2250$$

diff $\frac{dR}{dt}$

$$300t - 2250 = -150$$

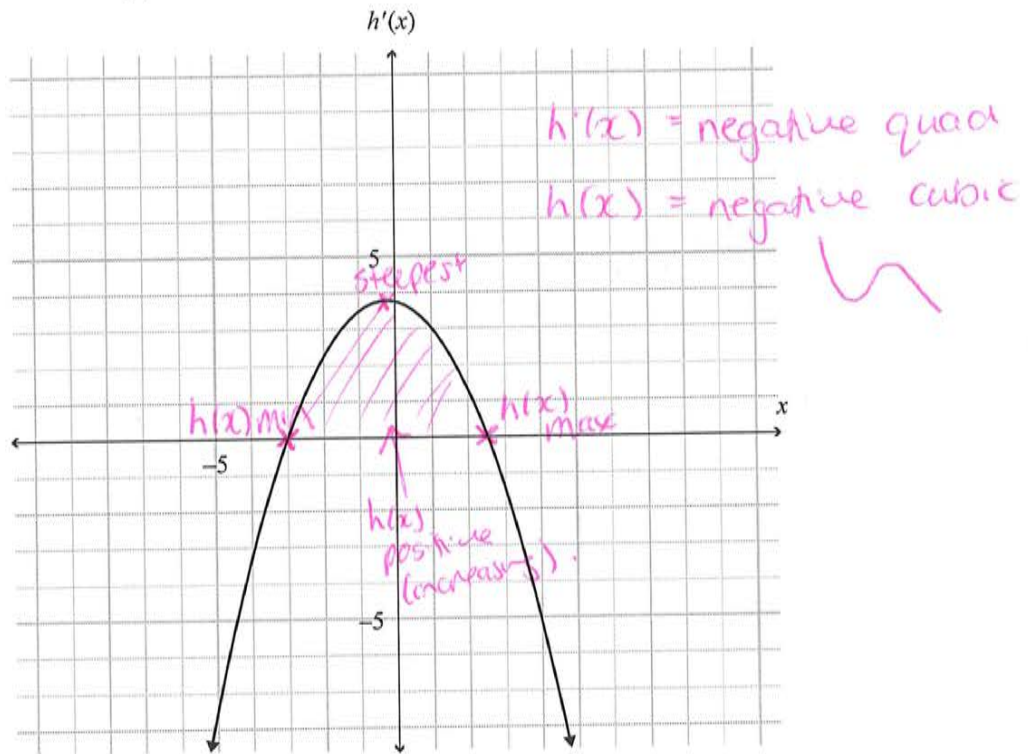
← (a) derivative found
AND set to -150

$$300t = 2100$$

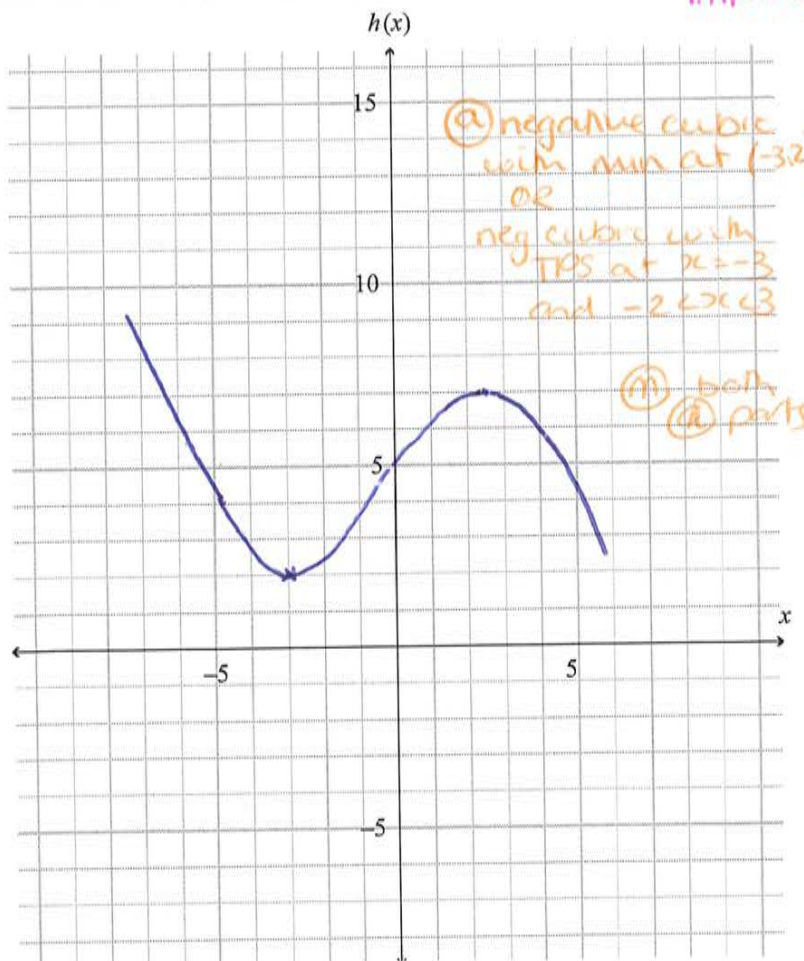
$$t = 7 \text{ years} \quad (m) \text{ } t \text{ found}$$

- (d) A gradient function $h'(x)$ is shown on the axes below.

ASSESSOR'S
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Sketch the function $h(x)$ if $h(x)$ passes through the point $(-3, 2)$. ← quite important!



If you need to redraw this graph, use the grid on page 17

$+C$

9

neg quad so y must be neg cubic
(y = -4)

ASSESSOR'S
USE ONLY

- (e) The gradient of the graph of a function is given by $\frac{dy}{dx} = 9x - 3x^2$.

At the maximum turning point of the graph of the function, $y = -4$.

Find the equation of the graph.

Note: The turning point where $y = -4$ must be shown to be a maximum.

grad $\rightarrow$ func
ant-diff

$$y = \frac{9x^2}{2} - x^3 + C$$

TP of cubic $\rightarrow \frac{dy}{dx} = 0$

$$9x - 3x^2 = 0$$

$$3x(3 - x) = 0$$

$$x = 0 \text{ or } x = 3$$

(a)

x-values
of TPs
OK

justify $x = 3$ is max.

① shape of neg cubic.

y is neg cubic

so second TP ($x = 3$) must be the max.

hence function passes through (3, -4)

$$y = \frac{9x^2}{2} - x^3 + C$$

$$-4 = \frac{9(3)^2}{2} - (3)^3 + C$$

$$-4 = \frac{27}{2} + C$$

$$C = -\frac{35}{2}$$

or ② $\frac{dy}{dx}$ is neg quad.

$\frac{dy}{dx}$ is a neg quad

y is decreasing $\rightarrow$ 0 $\leftarrow$ y is increasing
y is decreasing $\rightarrow$ 3 $\leftarrow$ y is decreasing.

\ / \ /

So $x = 3$ must be maximum.

(m) max TP identified as $x = 3$ AND justified.

or ③ 2nd derivative test

$$\frac{d^2y}{dx^2} = 9 - 6x$$

$$\text{when } x = 3 \quad \frac{d^2y}{dx^2} = 9 - 6(3) < 0 \quad \therefore \text{maximum.}$$

④ or correct ant-diff set equal to -4
 $-4 = 4.5x^2 - x^3 + C$

- (f) The tangent to the graph of the function $y = px^2 + 4x - 5$ at the point where $x = 3$ passes through the point $(0, 4)$.

Find the value of p , and hence the equation of the tangent.

$$\frac{dy}{dx} = 2px + 4$$

when $x = 3$ $y = p(3)^2 + 4(3) - 5$

$$y = 9p + 7$$

$$\frac{dy}{dx} = 2p(3) + 4$$

$$= 6p + 4$$

$$y - y_1 = m(x - x_1)$$

① expression found for gradient when $x = 3$.

$$y - (9p + 7) = (6p + 4)(x - 3)$$

$$y - 9p - 7 = 6px - 18p + 4x - 12$$

goes through point $(0, 4)$

$$4 - 9p - 7 = 6p(0) - 18p + 4(0) - 12$$

$$-9p + 18p = 7 - 4 - 12$$

$$9p = -9$$

$$p = -1$$

② found p

$$y - (9(-1) + 7) = (6(-1) + 4)(x - 3)$$

$$y + 9 - 7 = -2x + 6$$

$$y + 2 = -2x + 6$$

$$y = -2x + 4$$

③ eqn of tgt found.



QUESTION THREE

- (a) Find the x -coordinate of the point on the graph of the function $f(x) = 4x - x^2$ where the gradient is equal to 10.

$$f'(x) = 4 - 2x \stackrel{\text{diff}}{=} 10 \quad \textcircled{a}$$

$$4 - 10 = 2x$$

$$-3 = x \rightarrow x = -3 \quad \textcircled{a}$$

gradient function
AND solution

- (b) A swimming pool is being filled.

At the deepest point, the depth of the water in the pool at any instant is h m.

The volume of water in the pool $V \text{ m}^3$ is given by

$$V(h) = 20h^2 + 40h$$

Find the rate at which the volume of water is changing with respect to the depth when the water in the pool is 0.75 m deep.

$$V'(h) = 40h + 40 \quad \textcircled{a}$$

$$V'(0.75) = 40(0.75) + 40$$

$$= 70 \text{ m}^3/\text{m} \quad \textcircled{a}$$

derivative found
AND rate of change

- (c) A balloon is being inflated with helium.

The volume $V \text{ cm}^3$ of helium in the balloon is given by $V = \frac{4}{3}\pi r^3$, where r cm is the radius of the balloon.

Find the rate at which the volume of gas in the balloon is changing with respect to the radius when the volume is $288\pi \text{ cm}^3$.

$$\frac{dV}{dr} = 4\pi r^2$$

$$\text{When } V = 288\pi$$

$$288\pi = \frac{4}{3}\pi r^3$$

$$r^3 = 216$$

$$r = 6$$

$$\text{when } r = 6$$

$$\frac{dV}{dr} = 4\pi \times 6^2$$

$$= 144\pi \text{ cm}^3/\text{cm}$$

rate of change found

derivative found
AND almost r .

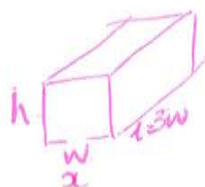


(d) The length of a cuboid is three times its width.

✦ The sum of the height, width, and length is 150 cm.

The volume $V \text{ cm}^3$ can be expressed as

$$V = 450x^2 - 12x^3 \quad \text{where the width is } x \text{ cm.}$$



Find the height of the cuboid for which the volume is maximum.

$$h + w + 3w = 150 \quad x = w.$$

$$h = -4x + 150$$

$$V = 450x^2 - 12x^3$$

$$V'(x) = 900x - 36x^2 = 0$$

$$x(900 - 36x) = 0$$

$$x = 0 \text{ or } 36x = 900$$

$$x = 25$$

Ⓐ derivative found
and set to 0.
AND x found.

max volume when $x = 25$

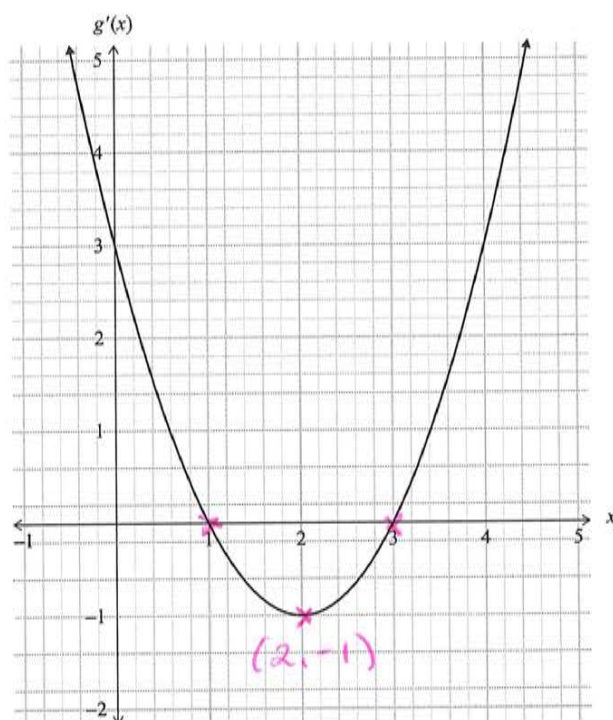
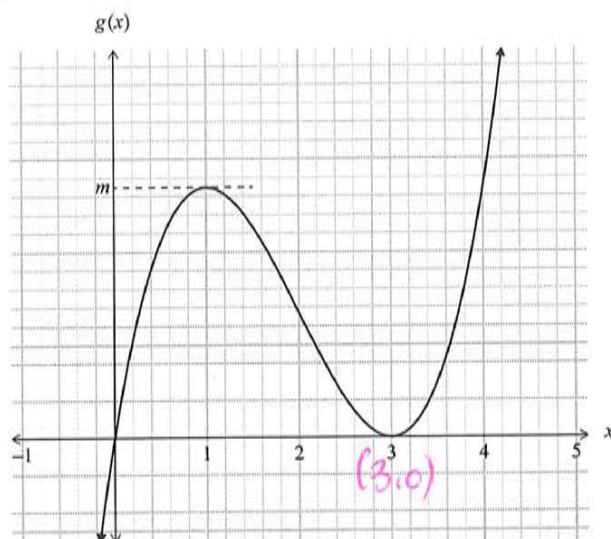
$$h = -4(25) + 150$$

$$h = -100 + 150$$

$$h = 50 \text{ cm}$$

Ⓜ h found.

- (e) The graph of a function $g(x)$ and the graph of its gradient function, are shown below.



Find the value m , the y -value for the maximum turning point of the function $g(x)$.

$$g'(x) = A(x-1)(x-3)$$

using $(2, -1)$ to solve
for A :

$$-1 = A(2-1)(2-3)$$

$$-1 = A \times 1 \times -1$$

$$-1 = -A$$

$$A = 1$$

$$g'(x) = (x-1)(x-3)$$

$$g'(x) = x^2 - 4x + 3$$

$$g(x) = \frac{x^3}{3} - 2x^2 + 3x + C$$

+C

using (3,0) to solve for C

$$0 = \frac{(3)^3}{3} - 2(3)^2 + 3(3) + C$$

$$0 = 0 + C$$

$$C = 0$$

$$g(x) = \frac{x^3}{3} - 2x^2 + 3x$$

(m) $g(x)$ found
with evidence $C=0$

when $x=1$,
 $g(x) = \frac{1}{3} - 2 + 3$

$$g(x) = \frac{4}{3} \quad \text{m found.}$$

- (f) An aircraft is travelling at 70 m s^{-1} when it lands.
Its speed changes at a constant rate of -3.3 m s^{-2} .



Use calculus to find how far the aircraft will travel from where it lands to where it has a speed of 4 m s^{-1} .

$$a(t) = -3.3$$

$$v(t) = -3.3t + C$$

using $t=0, v=70$ to find C.

$$70 = -3.3(0) + C$$

$$C = 70$$

$$v(t) = -3.3t + 70 \quad \text{(a)}$$

$$s(t) = -\frac{3.3t^2}{2} + 70t + C$$

using $t=0, s=0$ to find C

$$0 = 0 + 0 + C$$

$$C = 0$$

$$s(t) = -\frac{3.3t^2}{2} + 70t \quad \text{(m)}$$

$$v(t) = 4 = -3.3t + 70$$

$$-66 = -3.3t$$

$$t = 20$$

$$\text{when } v=4, t=20$$

$$s(20) = -\frac{3.3(20)^2}{2} + 70(20)$$

$$= 740 \text{ m} \quad \text{(e)}$$